

Test and Reliability Exercise Sheet 5

Important: Please fill out the online questionnaire for the course evaluation until July 12th. We would very much appreciate it!

Exercise 1

Consider the implication graph in Figure 1 with current decision level 6.

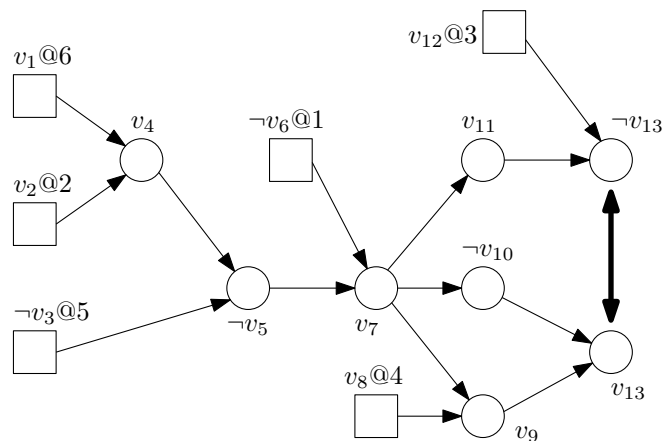


Figure 1: Implication graph

- Which UIPs are in the implication graph?
- Build a set of clauses that makes a DPLL solver produce this implication graph.
- Compute the conflict clause for the 1UIP.

Exercise 2

- Consider a circuit with five inputs a , b , c , d and e and three outputs O_1 , O_2 and O_3 . Output O_1 depends on a , b , c and d ; output O_2 depends on a , c and e and output O_3 depends on b , d and e . Applying pseudo-exhaustive testing, state a partition with a minimal number of test signal inputs p (cf. Slide 5:18).

- b) State a minimal example for the general problem given in part a), for which the number of test signals p is larger than the number of inputs on which any output depends w . “Minimal” means: with a minimum number of inputs I and outputs O (there is neither an example with less inputs nor an example with less outputs). Show that your solution is indeed minimal.

Exercise 3

In the lecture an LFSR is presented which produces a sequence of maximal length $2^n - 1$ for $n = 3$, where n is the number of flipflops in the LFSR. State an LFSR with a maximal sequence length for $n = 4$, i.e. a sequence length of $2^4 - 1 = 15$. Therefore specify whether an XOR-operation with the value Q_n is performed between two flipflops and draw the LFSR. State also an initial value and the complete sequence.

Reminder: As mentioned in the lecture, the only combination that will never occur in a maximal sequence is the “all-zero” combination: 0000.

Exercise 4

- a) Are the polynomials $x^2 + 1$ and $x^2 - 1$ irreducible? These polynomials are over GF(2) (Galois Field with 2 elements), therefore $+$ is equal to XOR.
- b) Provide irreducible polynomials of degree 3 and 4.
- c) Every polynomial can be uniquely factorized into irreducible polynomials; similarly to the factorization of natural numbers into prime numbers.

What is the largest common divisor of $x^4 - 1$ and $x^4 - x^2$?

Deadline: Discussion on Thursday, July 9th, 2026